

Work Summary on the Use of Coupler for the Purpose of Polarimeter

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1 Fundamental Test & Measurement

1.1 Measure the Polarization State of Laser Using Power Meter at Different Powers

Objective: Investigate whether different power levels affect the polarization state.

Setup: The laser machine used is the LASER SOURCE MODEL S1FC780PM 780nm. The polarization state of the laser is measured using a power meter at different power levels (0.03mW, 0.09mW, 0.20mW).

Conclusion: The LASER SOURCE MODEL S1FC780PM 780nm at lower power is not purely linear but presents more circular components of the polarization state. The polarization state becomes more linear when the laser is at high power. At power levels higher than 0.20mW, it can be considered purely linear.

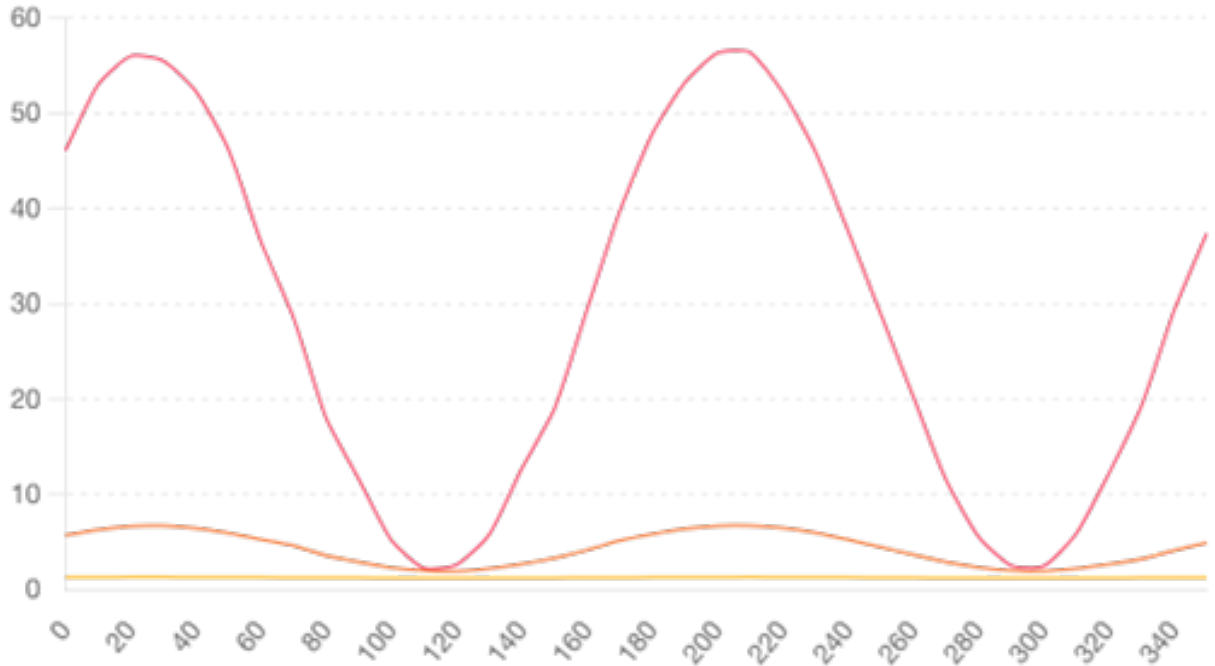


Figure 1: Polarization states at different power levels: 0.20mW (red), 0.09mW (orange), and 0.02mW (yellow). Y-axis: power in μW , X-axis: degrees.

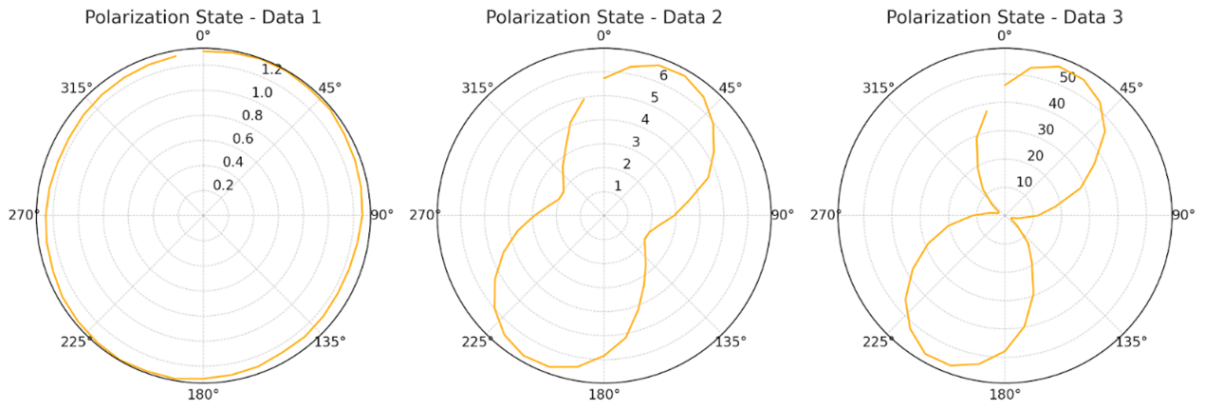


Figure 2: Polar form of the three power functions: 0.02mW, 0.09mW, and 0.20mW.

1.2 Relationship Between Polarization State Reading and Optical Fiber Angle

Objective: Investigate the relationship between the polarization state reading on the polarizer and the angle indicator of the optical fiber.

Setup: The polarizer used is THORSLABS CRM1/M. The power is set to 1mW.

Angle of Optical Fiber (degrees)	Polarization Angle (degrees)	Max Power (μ W)
0	100	108
30	70	108
60	40	108
90	10	108
120	340	108

Table 1: Relationship between optical fiber angle and polarization angle.

Conclusion: The sum of the angle of the optical fiber and the polarization angle is $100^\circ (+n \times 360^\circ)$. There is an 80-degree shift between the real polarization angle and the reading on the polarizer. When the polarizer shows zero, the real polarization state is at 280 degrees.

2 Investigation of Traditional Coupler (Two Waveguides Coupling)

2.1 Theory Model(Coupled Mode Theory)

Objective: Investigate the codirectional coupler presenting only the LP01 mode, which is the lowest order solution of the Bessel's equation of the wave differential equation when the input light is linearly polarized.

Assumptions:

- Weakly guiding for close values of cladding and core refractive index. Hence the LP approximation will be valid.
- Zero butt coupling coefficient and zero change in propagation constant, meaning modes in the individual waveguides are unperturbed.
- Conservation of power.
- Identical, lossless, and parallel waveguides.
- V number below 2.406 for specific wavelength.

Mode Expression: The LP01 single mode can be expressed as the weighted sum of the TE and TM modes (the TE and TM modes are the definition in slab waveguide instead of circular waveguide because of LP approximation), or more rigorously, the x-direction mode and y-direction mode.

2.1.1 Modes in Waveguides

The electric and magnetic fields in waveguides can be expressed as:

$$\mathbf{E} = \mathbf{E}_0(x, y)e^{-j\beta z} \quad (1)$$

$$\mathbf{H} = \mathbf{H}_0(x, y)e^{j\beta z} \quad (2)$$

When two waveguides (WGs) are close in proximity, they become coupled and exchange power periodically. Perturbation analysis includes assumptions that supermodes are the linear combination of individual guided modes[1]. However, each mode may be deformed in reality:

$$\mathbf{E} = A(z)\mathbf{E}_1 + B(z)\mathbf{E}_2 \quad (3)$$

$$\mathbf{H} = A(z)\mathbf{H}_1 + B(z)\mathbf{H}_2 \quad (4)$$

Where:

- 1st mode WG1
- 2nd mode WG2

To get general coupled mode theory equations, Maxwell's equations are needed. But coupling coefficient and change in propagation constant are normally ignored.

2.1.2 Mode Coupling Coefficient

$$k_{pq} = \frac{\omega\epsilon_0 \int \int (\epsilon_r - \epsilon_{q0}) \mathbf{E}_p^* \cdot \mathbf{E}_q dx dy}{2 \int \int \mathbf{E}_p^* \cdot \mathbf{H}_p + \mathbf{E}_p \times \mathbf{H}_p^* dz} \quad (5)$$

Measure of power leaks from WG(p) to WG(q) due to behavior of supermode [2].

For the HE₁₁ mode, we only need one:

$$k = \frac{\sqrt{\Delta}}{a} \frac{u^2}{v^3 K_1^2(\omega)} \sqrt{\frac{\pi a}{\omega D}} e^{-\frac{\omega}{a} D} \quad (6)$$

Where:

$D \rightarrow$ separation of core centers

$a \rightarrow$ core radius

$\Delta \rightarrow \frac{n_1^2 - n_2^2}{2n_1^2}$, n_1 and n_2 are core and cladding refractive indices

$v \rightarrow \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2}$ (dimensionless)

$u \rightarrow a \sqrt{K_1^2 n_1^2 - \beta^2}$

$\omega \rightarrow a \sqrt{\beta^2 - K_1^2 n_2^2}$

Explanation of terms:

- $\frac{\sqrt{\Delta}}{a}$ shows the refractive index effect.
- $\frac{u^2}{v^3 K_1^2(\omega)}$ shows the effect of modes, wavelength λ , and propagation constant.
- $\sqrt{\frac{\pi a}{\omega D}}$ is related to the core radius.
- $e^{-\frac{\omega}{a} D}$ represents the core separation.

However, once we separate the coupling coefficient to x-mode and y-mode, equation 6 can not predict k_x and k_y as they are not simply addition relationship.

2.1.3 Final Coupled Mode Equations

The final coupled mode equations are[3]:

$$\frac{dA}{dz} = -jk_a B \beta_1 e^{j\delta z} \quad (7)$$

$$\frac{dB}{dz} = -jk_b A \beta_2 e^{-j\delta z} \quad (8)$$

Where k_a and k_b are defined by $k_{21}, k_{12}, C_{12}, C_{21}, X_1, X_2$. For my project, simple assumptions are made $\delta = 0$, $\gamma = k$, $F = 1$, $k_a = k_b = k_{12} = k_{21}$

2.1.4 Codirectional Coupling and Power Derivation

For power injected to only one waveguide with initial condition $A(0) = A_0$, $B(0) = 0$:

$$A(z) = A_0 \left[\cos(\gamma z) - j \frac{\delta}{\gamma} \sin(\gamma z) \right] e^{-j\delta z} \quad (9)$$

$$B(z) = -A_0 \frac{jk}{\gamma} \sin(\gamma z) e^{j\delta z} \quad (10)$$

Where:

$$\tilde{P}_a(z) = \frac{|A(z)|^2}{|A_0|^2} = 1 - F \sin^2(\gamma z) \quad (11)$$

$$\tilde{P}_b(z) = \frac{|B(z)|^2}{|A_0|^2} = F \sin^2(\gamma z) \quad (12)$$

And:

$$F = \left(\frac{k}{\gamma} \right)^2 = \frac{1}{1 + \left(\frac{\delta}{k} \right)^2} \quad (13)$$

2.1.5 Polarization dependence of Power and Splitting ratio

The expressions for P_A and P_B of Single Mode LP01 can be estimated as the weighted sum of TE and TM modes(or more appropriately, the x-direction mode and y-direction mode)[4]:

$$P_A = P_x \cos^2 \theta + P_y \sin^2 \theta \quad (14)$$

$$(15)$$

Where: $\cos^2 \theta$ and $\sin^2 \theta$ are the weighting of x and y modes

P_x and P_y are both from (11) but with different coupling coefficient(x and y direction):

$$P_x = \cos^2 \left(\frac{k_x}{2} L_0 \right) \quad (16)$$

$$P_y = \cos^2 \left(\frac{k_y}{2} L_0 \right) \quad (17)$$

$$(18)$$

Where: γ has been replaced with $\frac{k}{2}$ instead of k , and let z equals the interaction length L_0

Hence P_A will be:

$$P_A = \cos^2 \left(\frac{k_x}{2} L_0 \right) \cos^2 \theta + \cos^2 \left(\frac{k_y}{2} L_0 \right) \sin^2 \theta \quad (19)$$

The same thing happens for waveguide B from (12):

$$P_B = P'_x \cos^2 \theta + P'_y \sin^2 \theta \quad (20)$$

$$P'_x = \sin^2 \left(\frac{k_x}{2} L_0 \right) \quad (21)$$

$$P'_y = \sin^2 \left(\frac{k_y}{2} L_0 \right) \quad (22)$$

$$P_B = \sin^2 \left(\frac{k_x}{2} L_0 \right) \cos^2 \theta + \sin^2 \left(\frac{k_y}{2} L_0 \right) \sin^2 \theta \quad (23)$$

The splitting ratio r is defined as:

$$r = \frac{P_B}{P_A + P_B} \quad (24)$$

$$= \frac{\sin^2\left(\frac{k_x}{2}L_0\right)\cos^2\theta + \sin^2\left(\frac{k_y}{2}L_0\right)\sin^2\theta}{\cos^2\theta\left(\sin^2\left(\frac{k_x}{2}L_0\right) + \cos^2\left(\frac{k_x}{2}L_0\right)\right) + \sin^2\theta\left(\sin^2\left(\frac{k_y}{2}L_0\right) + \cos^2\left(\frac{k_y}{2}L_0\right)\right)} \quad (25)$$

$$= \frac{\sin^2\left(\frac{k_x}{2}L_0\right)\cos^2\theta + \sin^2\left(\frac{k_y}{2}L_0\right)\sin^2\theta}{\cos^2\theta + \sin^2\theta} \quad (26)$$

$$= \sin^2\left(\frac{k_x}{2}L_0\right)\cos^2\theta + \sin^2\left(\frac{k_y}{2}L_0\right)\sin^2\theta \quad (27)$$

$$= \sin^2\left(\frac{k_x}{2}L_0\right) + \sin^2\theta\left(\sin^2\left(\frac{k_y}{2}L_0\right) - \sin^2\left(\frac{k_x}{2}L_0\right)\right) \quad (28)$$

Where: L_0 and θ are variables once we fix the separation of coupler, as well as x and y coupling coefficients. Moreover, the $\frac{k_y}{2}L_0$ is in radians while the θ is in degrees.

Plotting the 3D model, we expect to see polarization dependence of the splitting ratio, especially at certain interaction length with obvious changes. By looking at the theory model for longer range of interaction length, we could observe that the peaks and troughs of the splitting ratio curves for the different polarization state do not coincide but rather alternate. For long enough, it is absolutely possible for two polarization state have an approximate 90-degree phase difference relative to the interaction length, which has the same observation as [5] for DCs.

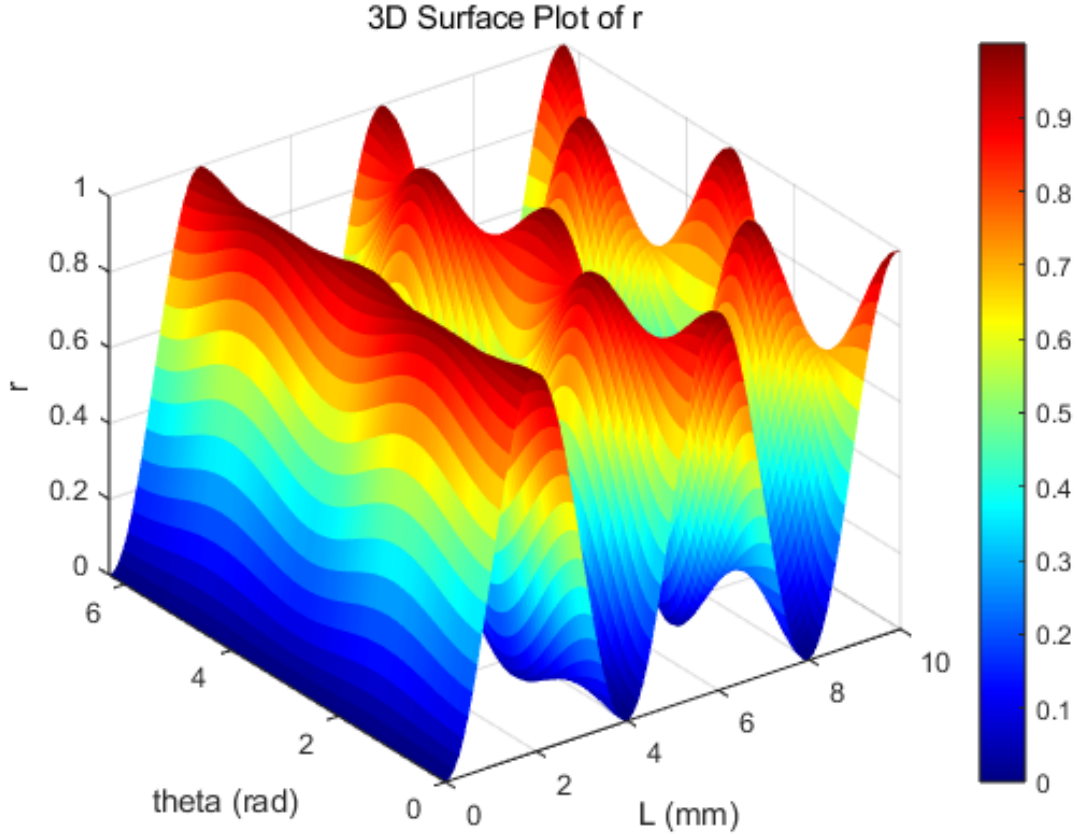


Figure 3: 3D Figure from the theory model for splitting ratio

Where: k_x and k_y used are 1.58mm^{-1} and 1.83mm^{-1} from experimental measurements

2.1.6 Polarimeter Model

From the splitting ratio's mathematical model, polarization angle θ could be calculated and predicted by measuring the splitting ratio if k_x , k_y and L_0 are all known and calibrated in advance. Hence the polarization state can be measured:

$$\theta = \arcsin \left(\sqrt{\frac{r - \sin^2 \left(\frac{k_x}{2} L_0 \right)}{\sin^2 \left(\frac{k_y}{2} L_0 \right) - \sin^2 \left(\frac{k_x}{2} L_0 \right)}} \right) \quad (29)$$

2.2 Femtosecond Fabrication of Coupler

2.2.1 Design of the Coupler

The designed schematic of the dimensions of directional coupler is the same as [6] except that samples with different separation have been fabricated. The separation distance we would like to investigate are 0.008mm, 0.006mm and 0.004mm.

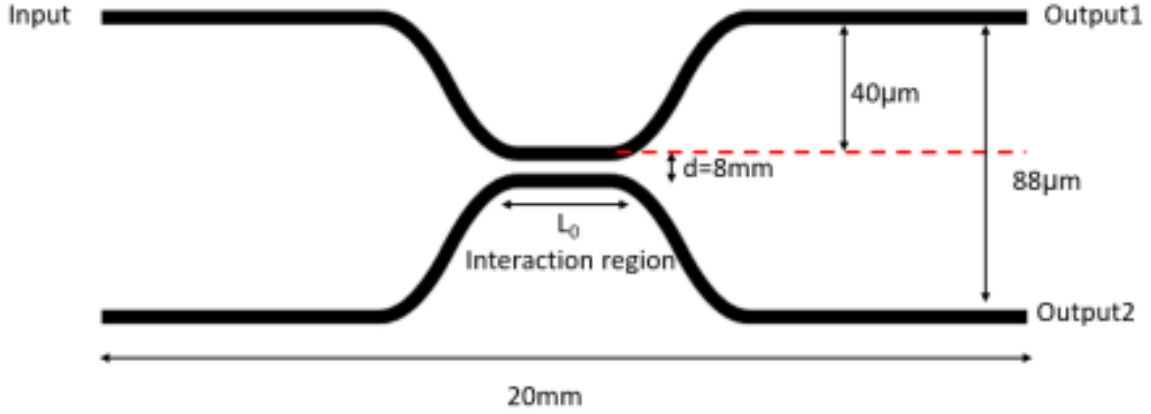


Figure 4: Schematic of dimension of traditional coupler

2.2.2 Fabrication Process

Fabrication Code The Aerotech G-code that controls the motion of the stage is generated by the MATLAB script which defines all the parameters of the traditional coupler and plot a line describing the motion of the laser path.

Fabrication System Same system setting in [6] and all the details can be found.

Experimental Procedure To prepare the borosilicate glass sample (Corning Eagle2000) for waveguide fabrication, a 20mm $\times$ 10mm $\times$ 1.1mm chip was mounted on a microscope slide to prevent movement. The surface was cleaned with acetone or ethanol to remove dust and particles. The sample was placed on a stage and focused in the z-direction using a CCD camera. Offsets in the 3 axes were noted and accounted for during fabrication.

For SLM calibration, the laser was focused on the sample surface, adjusting for an optimal Airy-like reflection shape by tuning the Zernike coefficients. Waveguide fabrication started outside the glass sample, with the pulse picker on, and the stage moved to write the waveguide uniformly across the sample. Post-fabrication, both ends were polished using a Krelltech Flex Waveguide Polisher with 30 μ m, 9 μ m, and 3 μ m pads consecutively for 5 minutes each to remove edge aberrations and enhance waveguide quality[6].

2.3 Experiment on Measuring Splitting ratio at different interaction length and polarization state with 0.008mm separation

2.3.1 Experiment Setup

The splitting ratio r is measured for interaction length 2mm-10mm and polarization state from 80 to 170 degrees that range over 90 degrees. The interaction length has 17 settings(2, 2.5, 3...9.5, 10), and polarization state has 7 settings(80, 95, 110...155, 170). Hence there are 119 sets of splitting ratio measured. The laser used was LASER SOURCE MODEL S1FC780PM 780nm and the power was set to 0.2mW.

The polarization state was changed by rotating the optical fiber as we already know the relationship between the polarization state and the optical fiber angle. For each polarization state, 17 couplers were measured and repeat over and over again after rotating the fiber.

The light intensity of two waveguides at the out ports were measured by the THORLABS PM100D. By subtracting the ambient intensity measured, the splitting ratio could be calculated.

2.3.2 Experimental Results

Table Value: The following table shows splitting ratio at different interaction length and polarization state(real and polarizer reading)

	2mm	2.5mm	3mm	3.5mm	4mm	4.5mm	5mm	5.5mm	6mm	6.5mm	7mm	7.5mm	8mm	8.5mm	9mm	9.5mm	10mm	interaction length	
80	0.154	0.133	0.34	0.56	0.557	0.591	0.448	0.193	0.2	0.295	0.558	0.672	0.613	0.297	0.224	0.209	0.487		0
95	0.17	0.146	0.316	0.518	0.667	0.519	0.462	0.182	0.288	0.37	0.561	0.547	0.522	0.409	0.317	0.313	0.417		15
110	0.196	0.194	0.304	0.522	0.609	0.596	0.471	0.278	0.213	0.306	0.533	0.551	0.565	0.404	0.286	0.326	0.408		30
125	0.243	0.198	0.29	0.48	0.577	0.609	0.517	0.292	0.303	0.323	0.476	0.536	0.528	0.473	0.421	0.345	0.383		45
140	0.281	0.17	0.267	0.393	0.5	0.619	0.589	0.423	0.327	0.277	0.38	0.417	0.469	0.559	0.512	0.415	0.38		60
155	0.349	0.196	0.143	0.355	0.452	0.703	0.676	0.485	0.389	0.286	0.265	0.396	0.426	0.571	0.547	0.438	0.4		75
170	0.333	0.158	0.159	0.31	0.385	0.711	0.689	0.561	0.343	0.25	0.3	0.368	0.439	0.618	0.588	0.517	0.389		90
real polarisation																			polarisation state on polarizer

Figure 5: Splitting Ratio Measurement Data

Splitting Ratio vs. Interaction Length for Each Polarization: For each polarization investigated, the best fit curve of splitting ratio is plotted. By observing the frequency, coupling coefficient can be estimated. It is further verified that k_x and k_y are nearly $1.58mm^{-1}$ and $1.83mm^{-1}$.

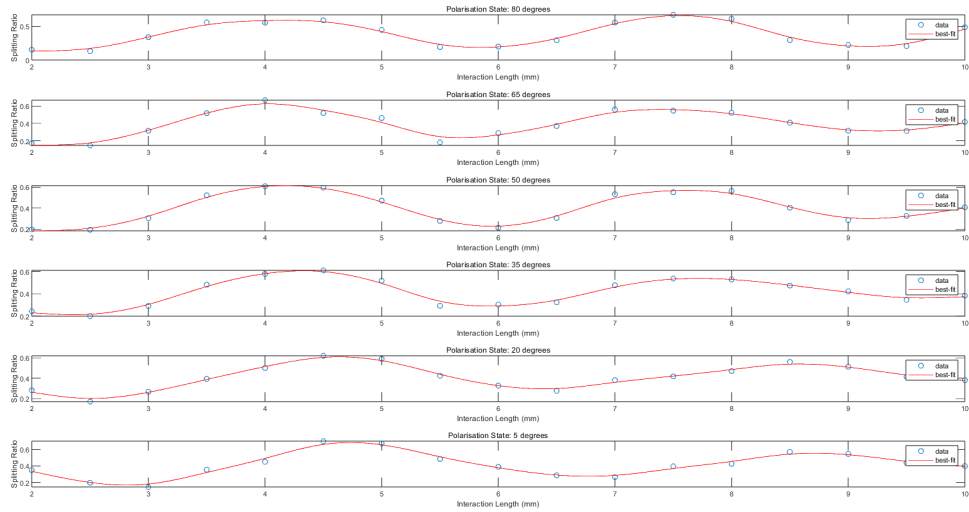
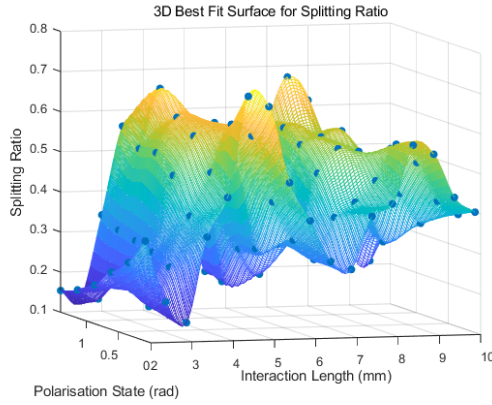
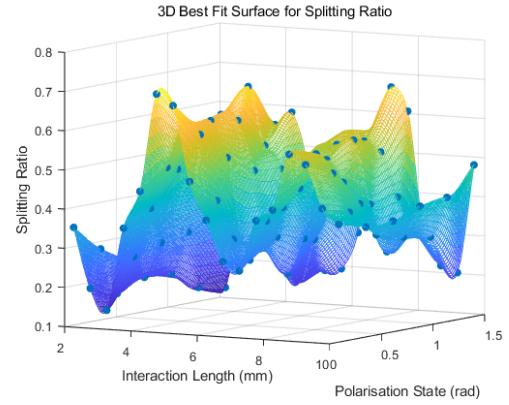


Figure 6: Splitting Ratio vs. Interaction Length

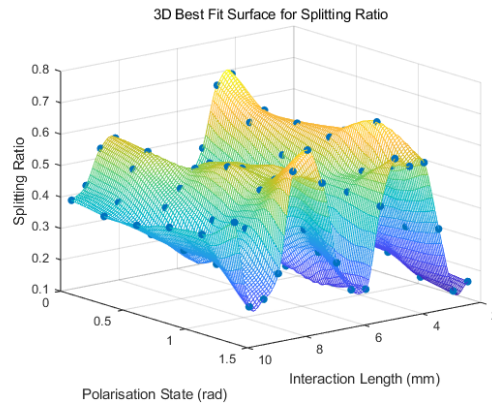
Comparison between Theory Model and Best-fit-surface: The following image matrix plots are the best-fit-surface plot for the experimental data from different angle, which shows two period of oscillation and polarization dependence at different interaction length.



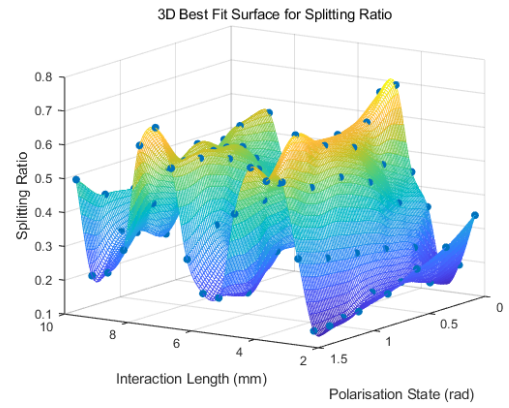
(a) Angle 1



(b) Angle 2

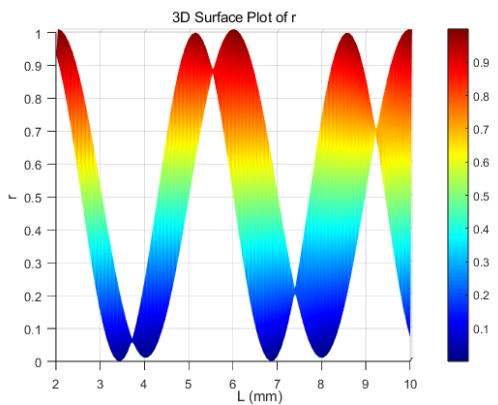


(c) Angle 3

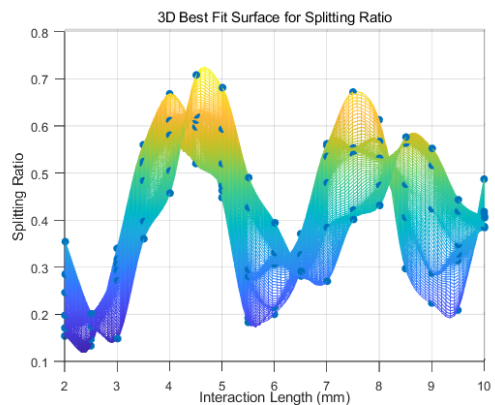


(d) Angle 4

Figure 7: 3D Figure Matrix from different angles



(a) Angle 1



(b) Angle 2

Figure 8: Side View Comparison between Theory Model and Best-fit

- Both the theoretical model and the best fit surface from the experimental data exhibit a periodic pattern in the splitting ratio (r) as a function of both the interaction length (L) and the polarization state (θ). The overall shape of the surfaces in both figures indicates a clear dependence on these two parameters, suggesting that the theoretical model captures the essential behavior observed in

the experiments.

- The amplitudes of the splitting ratio peaks in both the theoretical and experimental surfaces are comparable, though there are some discrepancies. The theoretical model predicts slightly higher peak values compared to the experimental best fit surface. These differences could be attributed to experimental imperfections, measurement errors, or assumptions made in the theoretical model that do not fully capture all physical phenomena present in the actual experiment. However, the main problem is that when we measure the power at out-ports, the power leakage to the background is also measured. Although the background looks black in the camera, it still has significant power. Hence if we could subtract the background power when we measure the power at waveguide outports, the results will be more clear, precise and perfect.

In summary, the comparison demonstrates that the theoretical model effectively describes the general behavior of the splitting ratio as a function of interaction length and polarization state. The similarities in shape, frequency, and trend between the two surfaces suggest that the theoretical model is robust, though minor discrepancies highlight potential areas for further investigation and model refinement.

Model Compensation: However, there are still some differences between the theoretical model and the experimental data. We need to investigate the reasons behind these differences and how they occur.

The first difference is in the magnitude. The theoretical model suggests that the splitting ratio can switch back and forth between 0 and 1, but this is not achievable in reality. The coupling process will not be perfect, resulting in a reduction in the splitting ratio.

The second difference lies in the shift of the interaction length. The experimental data would match the theoretical model more closely if the x-axis were shifted to the left. This discrepancy arises because the fabrication process cannot achieve perfect interaction lengths, leading to the waveguides potentially interacting earlier than expected. For example, the theoretical interaction length of 3 mm might correspond to an actual interaction length of 2 mm in the fabricated sample.

The third difference is the ambient power which originates from the measurements as zero measured power would not be possible at WG-B output because of the ambient noise. While the theoretical model predicts that the splitting ratio can drop to zero, the experimental measurements do not show this behavior due to environmental noise or ambient power. Therefore, it is necessary to compensate for this factor.

Through careful adjustments, the optimal results were found to be a 0.75 reduction in magnitude, a 1.16 mm shift in interaction length, and an ambient power of 0.05. Hence the modified splitting ratio r is

$$r = 0.75 * \left(\sin^2 \left(\frac{k_x}{2} (L_0 + 1.16) \right) + \sin^2 \theta \left(\sin^2 \left(\frac{k_y}{2} (L_0 + 1.16) \right) - \sin^2 \left(\frac{k_x}{2} (L_0 + 1.16) \right) \right) + 0.05 \right)$$

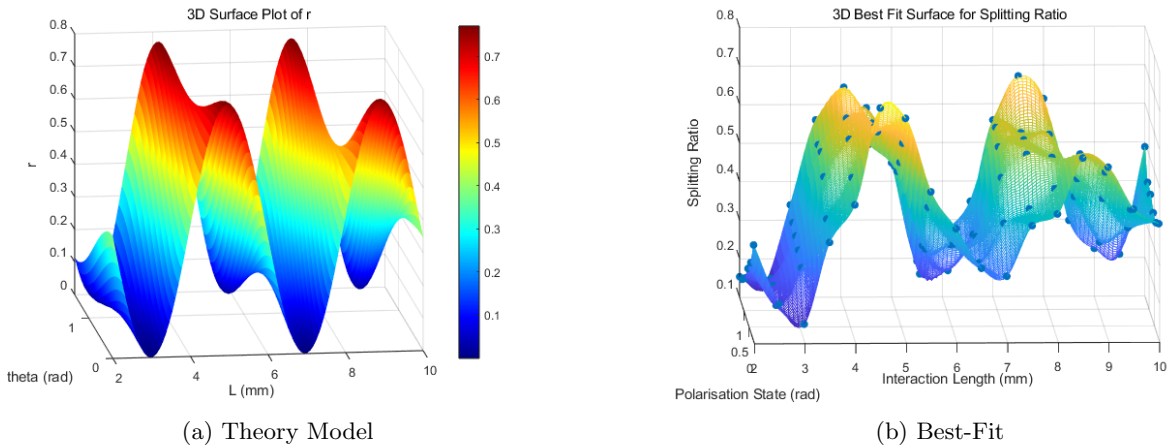


Figure 9: Comparison between Theory Model with compensation and Best-fit

After the compensation, the Mean Square Error(MSE) between the theory model and experimental data is 0.0129 which is nearly 1.3 percent.

Choosing Suitable Interaction Length for Polarimeter Use: The purpose of designing a polarimeter is to achieve significant changes in the splitting ratio with variations in the polarization state. This is not feasible at the crest or trough of the splitting ratio curve. Instead, significant changes are observed in the regions between the crest and trough. A 9mm interaction length exemplifies this behavior, making it a suitable candidate for the original type of polarimeter(for separation 0.008mm).

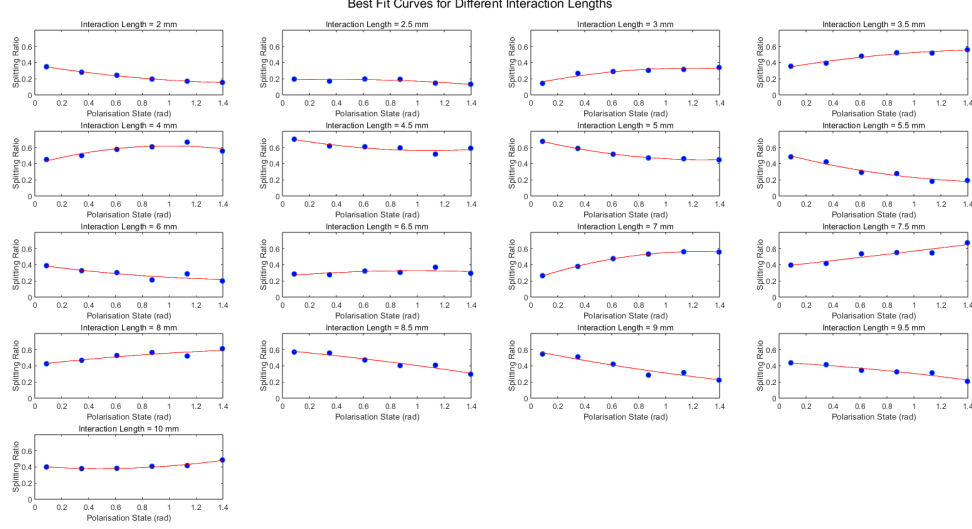


Figure 10: Splitting Ratio vs. polarization at different interaction length (Experiment)

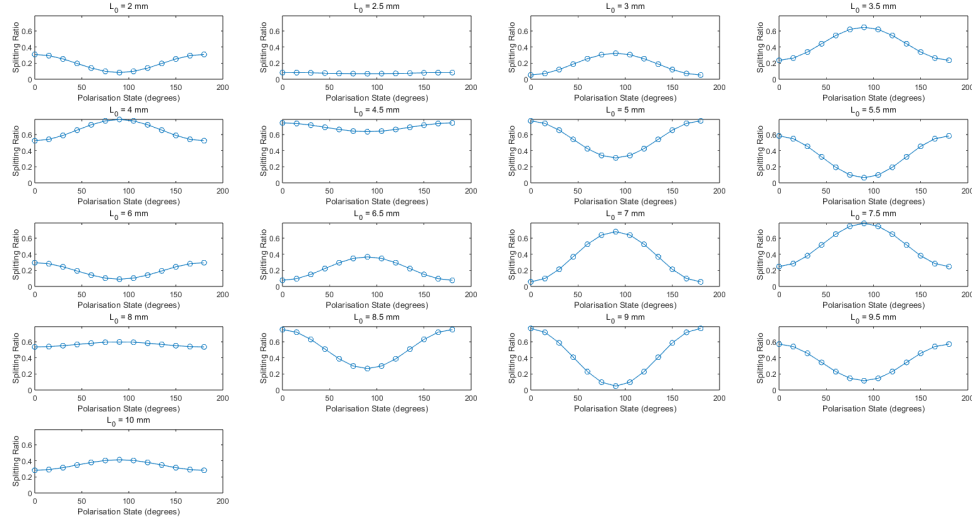


Figure 11: Splitting Ratio vs. polarization at different interaction length (Compensated Theory)

For further refinement, the appropriate interaction length for the polarimeter depends on various coupling coefficients, which are influenced by the separation length. This means that for different separation lengths, there are corresponding optimal interaction lengths. Understanding the relationship between them allows for easier structuring of measurement devices. For instance, when measuring light information in a waveguide, an adjacent empty waveguide with any separation and a certain interaction length can be added, minimizing the impact on the chip or optical system. This flexibility in interaction length simplifies and enhances the design.

2.4 Influence of different separation on coupling coefficient and suitable interaction length for polarization measurement

2.4.1 Estimation for Suitable Interaction Length Range before Fabrication

From Equation 6, we can roughly estimate how separation influences the coupling coefficients. This allows us to predict the behavior of the coupling coefficients in both the x and y directions, as well as to simulate the theoretical model. Consequently, we can determine the appropriate range of interaction lengths for fabrication for separation 0.004mm and 0.006mm.

$k = \frac{\sqrt{\Delta}}{a} \frac{u^2}{v^3 K_1^2(\omega)} \sqrt{\frac{\pi a}{\omega D}} e^{-\frac{\omega}{a} D}$ presents the influence of root and exponential power. The power term is hard to determine, we shall start with the root.

For separation 0.004mm, k_x and k_y are then calculated to be $2.234mm^{-1}$ and $2.588mm^{-1}$. By observing the 3D plot of the theory model, 2mm-6mm range is enough for investigating the periodic behavior. For separation 0.006mm, k_x and k_y are then calculated to be $1.824mm^{-1}$ and $2.113mm^{-1}$. 2mm-6mm is also enough for this case.

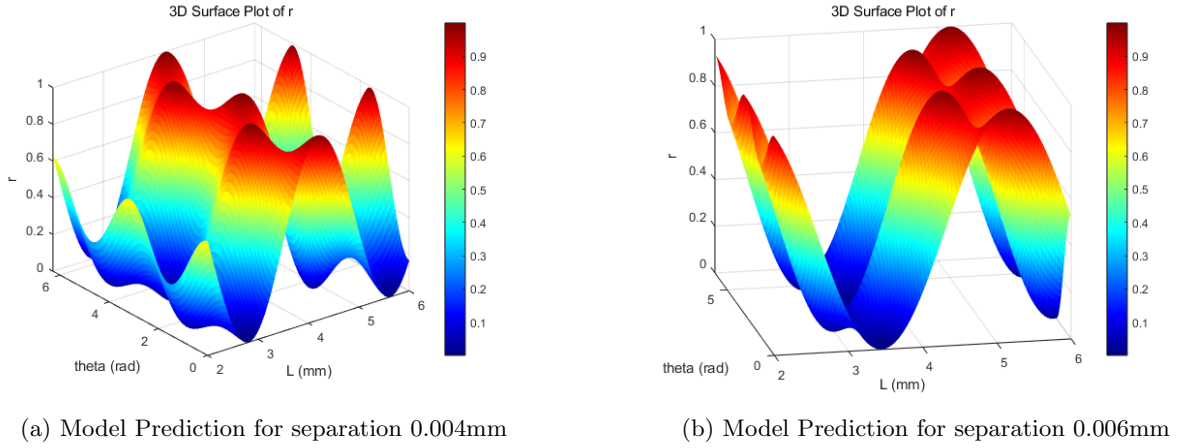


Figure 12: Model Prediction

However, it is also noteworthy that there is a range of interaction lengths, specifically 16mm to 18mm, that is polarization independent or insensitive. Polarization insensitivity in directional couplers can offer significant benefits and application to the quantum information processing. If we could also remain the polarization state, it would be extremely useful to the fidelity of quantum state, stability of quantum information system or simplification of the design of integrated photonic circuits. This might be a different techniques other than the discussion in [7]. To further investigate the validity of this observation, fabricating a coupler with this interaction length is worth considering.

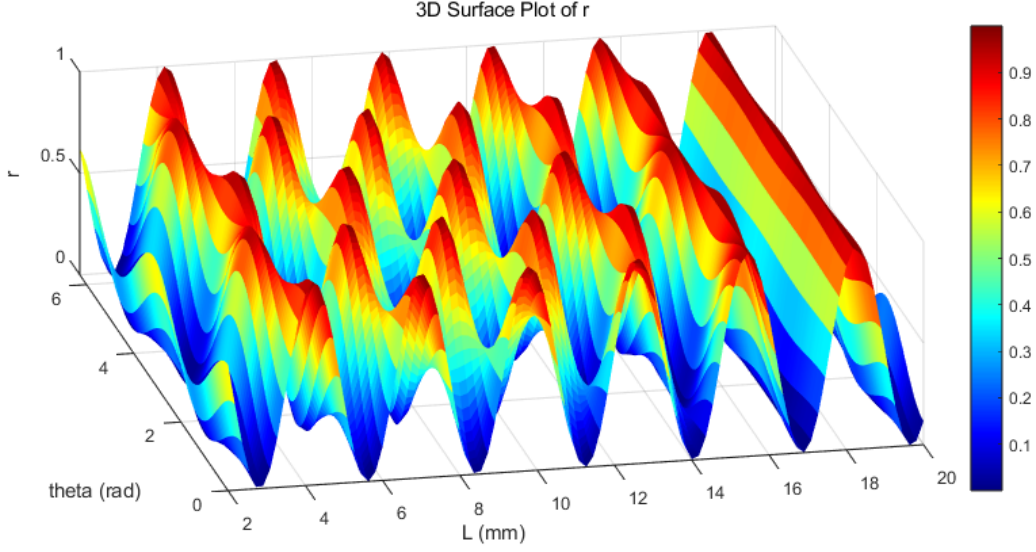


Figure 13: Model Prediction for separation 0.004mm with broader range

2.4.2 Experimental Results for 0.004mm Separation

Table Value: The following table shows splitting ratio at different interaction length and polarization state for 0.004mm separation distance.

separation =0.004mm	0 degree	15 degree	30 degree	45 degree	60 degree	75 degree	90 degree	polarisation state
2mm	0.3356	0.3060	0.4051	0.4118	0.4419	0.5522	0.5938	
2.5mm	0.6703	0.7286	0.5797	0.5417	0.4940	0.3944	0.2985	
3mm	0.3359	0.3497	0.3913	0.4348	0.4268	0.4699	0.4533	
3.5mm	0.6174	0.6212	0.6711	0.6512	0.6220	0.5926	0.5773	
4mm	0.3271	0.3534	0.3902	0.3614	0.3485	0.3684	0.3415	
4.5mm	0.5789	0.6522	0.4725	0.5902	0.5205	0.5581	0.5758	
5mm	0.5658	0.5417	0.3922	0.5238	0.4444	0.3696	0.3803	
5.5mm	0.5125	0.4167	0.5476	0.5652	0.5909	0.5918	0.5769	
6mm	0.5106	0.5146	0.5167	0.5200	0.5467	0.5571	0.5529	
Interaction Length (mm)								

Figure 14: Splitting Ratio Measurement Data for 0.004mm Separation

Splitting Ratio vs. Interaction Length for Each Polarization: For polarization states ranging from 0 to 90 degrees, the relationship between the splitting ratio and interaction length is plotted. The red line represents the best-fit curve, while the blue line indicates a rough fit. However, the best-fit function shows poor alignment with this data set. This discrepancy could be due to the separation distance between the waveguides being too small, resulting in unstable performance. k_x and k_y could be carefully estimated to be $6.16mm^{-1}$ and $4.189mm^{-1}$ by the following relationship[4].

$$\kappa_x = \frac{2\pi}{T_x} \quad (30)$$

$$\kappa_y = \frac{2\pi}{T_y} \quad (31)$$

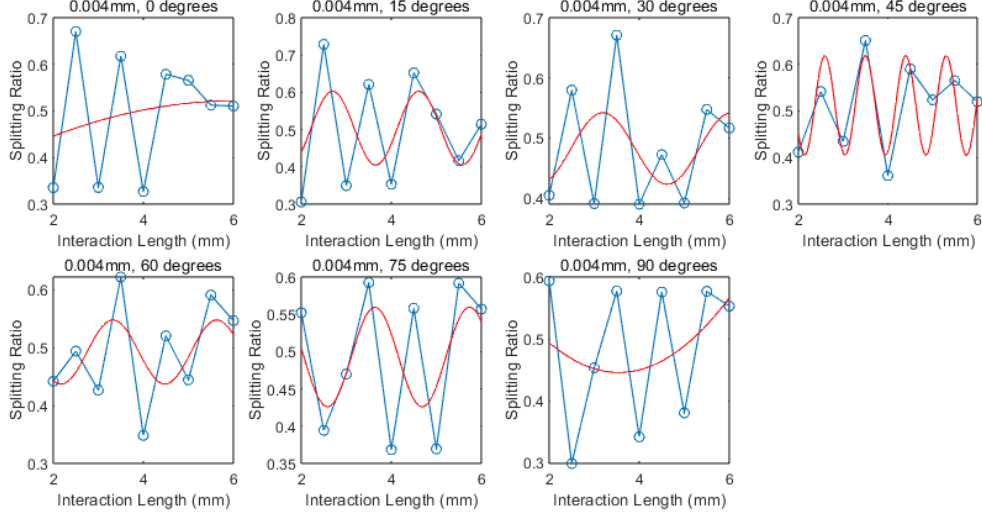


Figure 15: Splitting Ratio vs. Interaction Length for 0.004mm Separation

From observation, we do require more data points along the interaction length from 2mm-6mm, then we might obtain clear oscillation pattern.

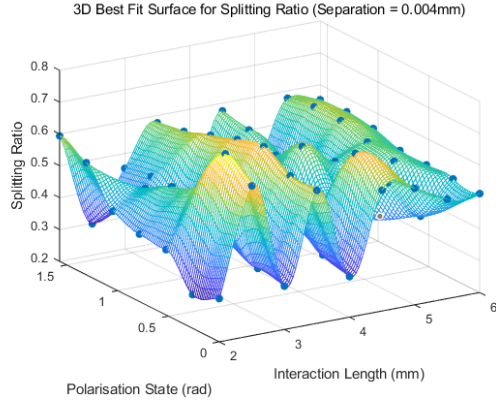
There are few possible reasons for the unstable performance when separation is too small:

- When the waveguide spacing is too small, it becomes sensitive to fabrication errors or imperfect laser-writing process which can significantly impact the coupling performance, resulting in actual performance deviating from the design values. For example, the shape of laser focal has slightly deformed as writing through the glass.
- Close waveguides creates stronger coupling effect creating complex coupling mode. Hence our assumption of weighted sum of TE and TM modes would break, as well as the predict model for splitting ratio. When the coupling gets more complicated, the use of sin/cos to make best-fit is no longer an appropriate choice.

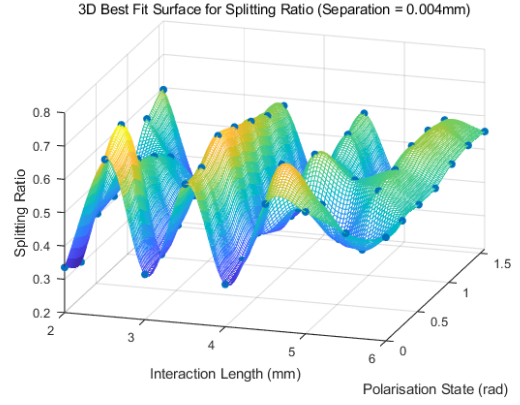
Moreover, compared with the predicted coupling coefficient, it shows great discrepancy ($6.16mm^{-1}$ with $2.234mm^{-1}$ and $4.189mm^{-1}$ with $2.234mm^{-1}$). The wrong estimation is caused by:

- Equation 6 is for single HE11 mode which can not be considered as sum weighted form of TE and TM as what we did for power. This simplification does not apply to coupling coefficient. That is also why the x coupling coefficient becomes even bigger than y coupling coefficient which will not happen if they change with the same scale when previously the x coupling coefficient is smaller than y coupling coefficient.
- The exponential power term has not been considered yet. It may have significant impact on the change of coupling coefficient.

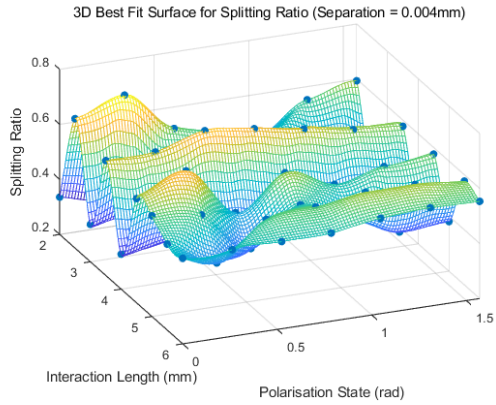
Comparison between Theory Model and Best-fit-surface: The following image matrix plots are the best-fit-surface plot for the experimental data from different angle, which shows at least three period of oscillation and polarization dependence behavior.



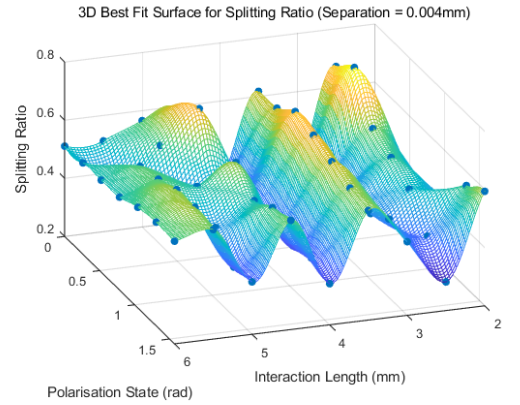
(a) Angle 1



(b) Angle 2



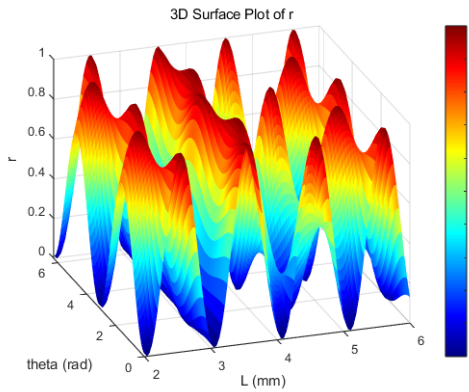
(c) Angle 3



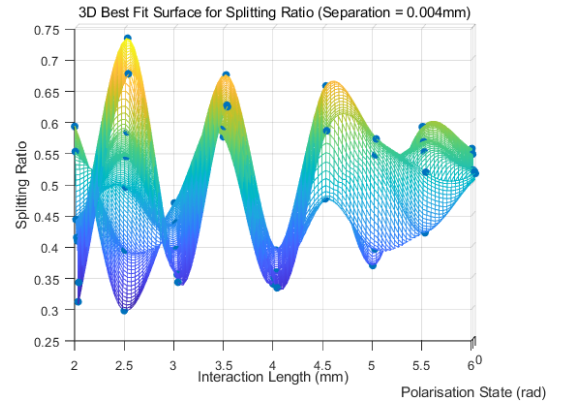
(d) Angle 4

Figure 16: 3D Figure from different angles

The theory model is plotted by applying the x and y coupling coefficient as 6.16mm^{-1} and 4.189mm^{-1} .



(a) Theory Model



(b) Best-Fit

Figure 17: Side View Comparison between Theory Model and Best-fit

Oscillatin Pattern Both the theoretical model and experimental results show four periods (or peaks) of oscillation within the 2mm-6mm range. From the side view plot, the measured results' periodic trend fundamentally matches the trend predicted by the theoretical model.

Aplitude Range The numerical range in the theoretical model spans from 0 to 1, while the numerical range of the experimental data spans from about 0.3 to 0.75, which is close to that of the theoretical model, indicating that the experimental data are on the same scale as the theoretical predictions. The

scatter observed is most likely due to power leakage caused by imperfectly fabricated waveguides and environmental noise.

MSE Minimization By observing the plot, the experiment shows good agreement with the theoretical prediction. However, we still need compensation for the assumptions made and the imperfect reality. Therefore, mean squared error (MSE) is needed to provide a quantified comparison between the two.

Model Compensation: In this case, the difference is not much obvious except the reduction in magnitude. As the trend and period match well, modification of the effective interaction length is not reasonable anymore. Hence we would more rely on the quantification method. To refine the theoretical model and better align it with the experimental data, we employed a constrained multi-parameter optimization method. The goal was to minimize the mean squared error (MSE) between the model predictions and the experimental results.

We used the L-BFGS-B algorithm[8], which is effective for optimizing parameters within specified bounds. The parameters optimized included the wave vector components k_x and k_y , the phase shift (*shift*), and the ambient power:

- k_x (5-7mm)
- k_y (3-5mm)
- *shift* (0-2mm)
- Ambient power (0-0.3 units)

The modified theoretical model including ambient power is given by:

$$r(L_0, \theta) = M \left(\sin^2 \left(\frac{k_x}{2} (L_0 + \text{shift}) \right) + \sin^2(\theta_{\text{rad}}) \left(\sin^2 \left(\frac{k_y}{2} (L_0 + \text{shift}) \right) - \sin^2 \left(\frac{k_x}{2} (L_0 + \text{shift}) \right) \right) \right) + \text{ambient power}$$

The optimization results are as follows:

- Optimal k_x : 5.9710
- Optimal k_y : 4.8987
- Optimal *shift*: 0.0
- Optimal ambient power: 0.1442
- Minimum MSE: 0.0167

Hence the model after optimization is:

$$r(L_0, \theta) = 0.75 \left(\sin^2 \left(\frac{k_x}{2} (L_0) \right) + \sin^2(\theta_{\text{rad}}) \left(\sin^2 \left(\frac{k_y}{2} (L_0) \right) - \sin^2 \left(\frac{k_x}{2} (L_0) \right) \right) \right) + 0.1442$$

where k_x and k_y are now 5.9710 mm^{-1} and 4.8987 mm^{-1}

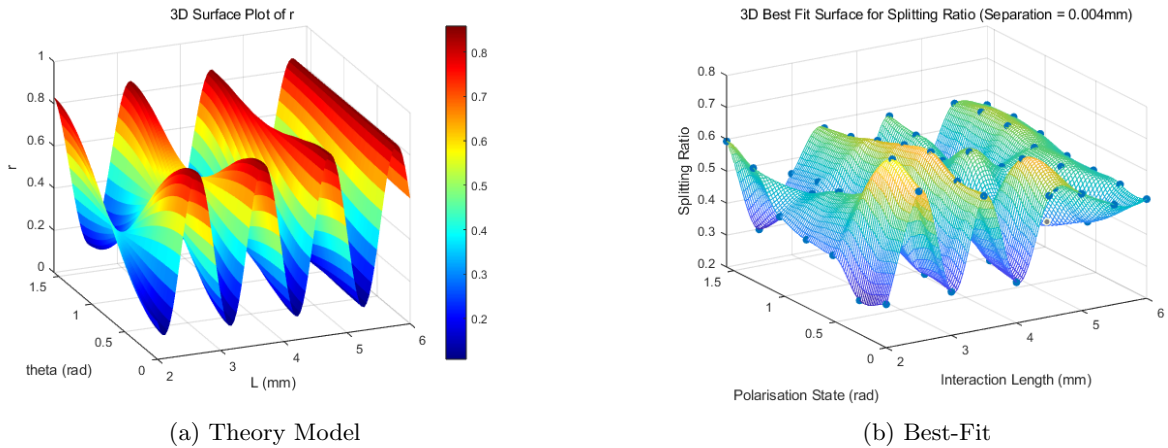


Figure 18: Comparison between Theory Model with compensation and Best-fit

MSE has been reduced to 0.0167 which is 1.67 percent.

Choosing Suitable Interaction Length for Polarimeter Use: For separation=0.004mm, the most suitable interaction length in the range 2mm-6mm is 2mm which has greatest slope and significant change with the variation of polarization state.

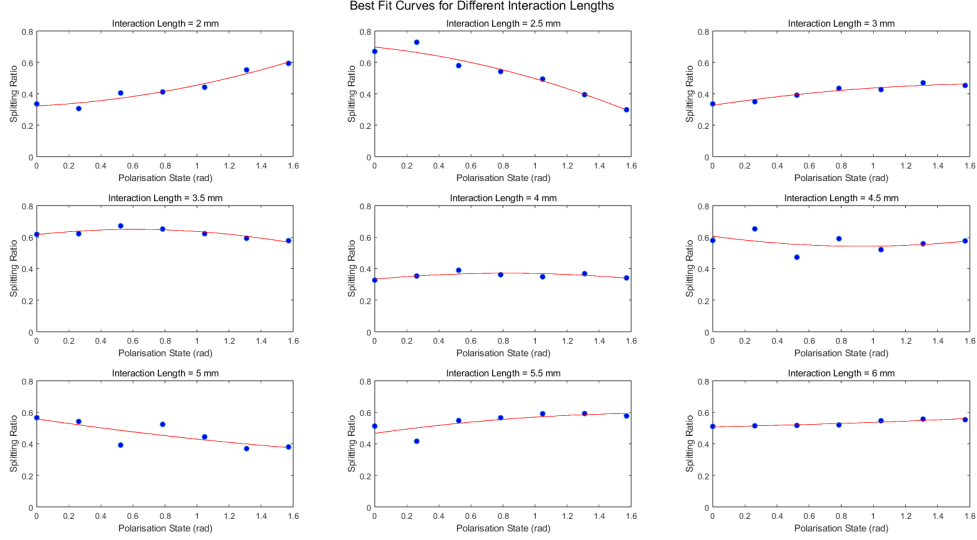


Figure 19: Splitting Ratio vs. polarization at different interaction length (Experiment)

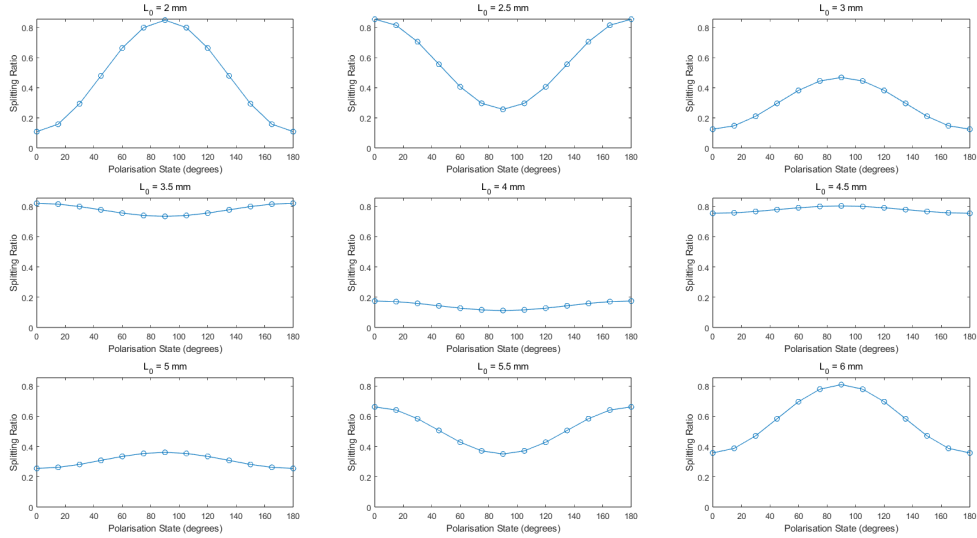


Figure 20: Splitting Ratio vs. polarization at different interaction length (Compensated Theory)

2.4.3 Experimental Results for 0.006mm Separation

Table Value: The following table shows splitting ratio at different interaction length and polarization state for 0.006mm separation distance.

separation =0.006mm	0 degree	15 degree	30 degree	45 degree	60 degree	75 degree	90 degree	polarisation state
2mm	0.5143	0.5299	0.4559	0.4658	0.4474	0.4098	0.3857	
2.5mm	0.7114	0.7519	0.7041	0.7614	0.7416	0.7027	0.7045	
3mm	0.6615	0.6990	0.6835	0.7093	0.6832	0.7534	0.8261	
3.5mm	0.2452	0.2414	0.3232	0.3333	0.3763	0.4343	0.4951	
4mm	0.2368	0.2033	0.2717	0.2762	0.2551	0.2391	0.1818	
4.5mm	0.7500	0.7963	0.8087	0.8087	0.7156	0.6100	0.5750	
5mm	0.8504	0.8254	0.7724	0.8587	0.7876	0.8235	0.7789	
5.5mm	0.2480	0.2391	0.2441	0.2778	0.3390	0.3978	0.4833	
6mm	0.2111	0.1613	0.2105	0.3253	0.4151	0.4400	0.3193	
Interaction Length (mm)								

Figure 21: Splitting Ratio Measurement Data for 0.006mm Separation

Splitting Ratio vs. Interaction Length for Each Polarization: For polarization states ranging from 0 to 90 degrees, the relationship between the splitting ratio and interaction length is plotted. The red line represents the best-fit curve, while the blue line indicates a rough fit. On the contrary, the best-fit function works well for this set of experimental data due to its regularity and stability. Hence k_x and k_y could be well calculated to be $3.307mm^{-1}$ and $3.11mm^{-1}$ (compared to previous estimation $1.824mm^{-1}$ and $2.113mm^{-1}$ which are wrong).

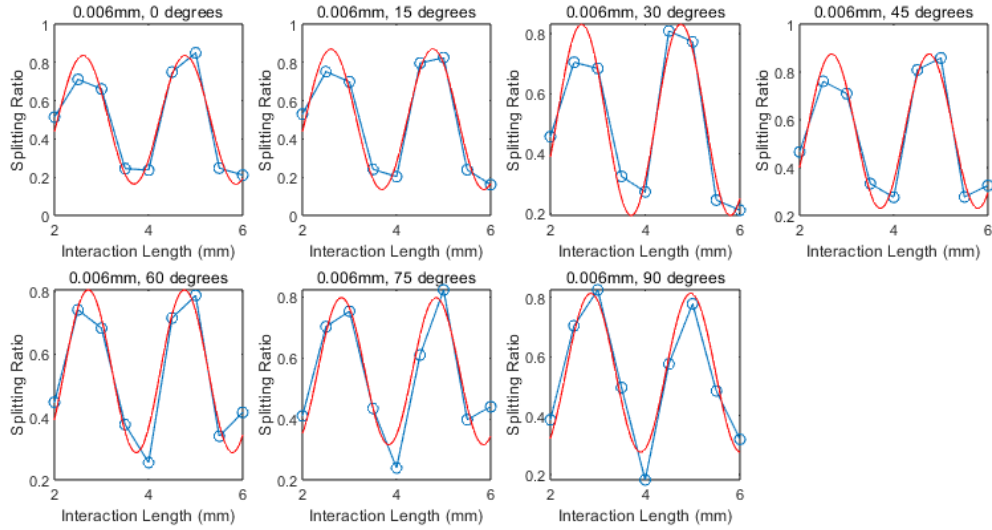
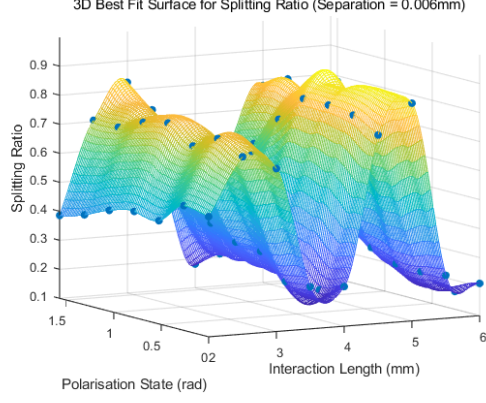
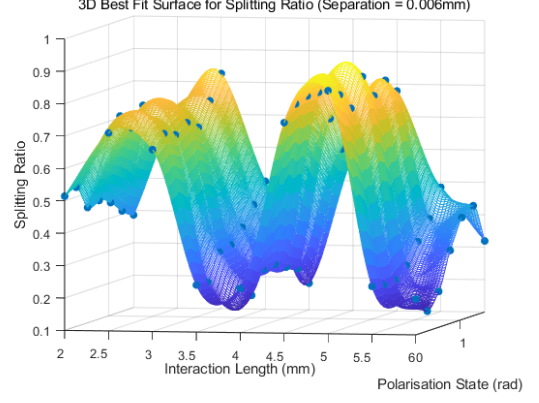


Figure 22: Splitting Ratio vs. Interaction Length for 0.006mm Separation

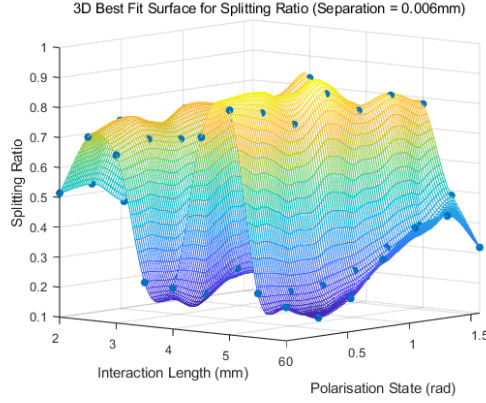
Comparison between Theory Model and Best-fit-surface: The following image matrix plots are the best-fit-surface plot for the experimental data from different angle, which shows at least two period of oscillation and polarization dependence behavior.



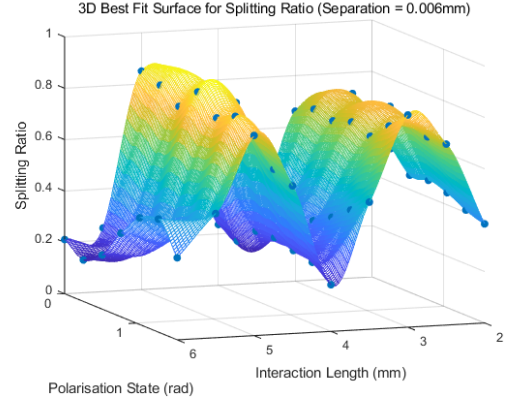
(a) Angle 1



(b) Angle 2



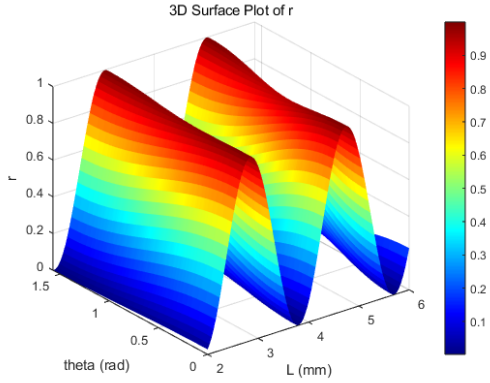
(c) Angle 3



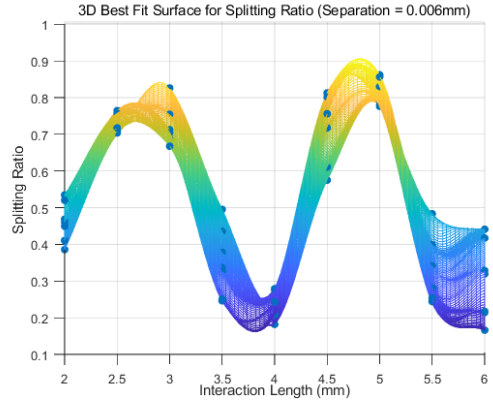
(d) Angle 4

Figure 23: 3D Figure from different angles

The theory model is plotted by applying the x and y coupling coefficient as 3.307mm^{-1} and 3.11mm^{-1} .



(a) Theory Model



(b) Best-Fit

Figure 24: Side View Comparison between Theory Model and Best-fit

The measured results are significantly similar to the model predictions, though there is a reduction in scale and a slight shift.

Oscillatin Pattern Both the theoretical model and experimental results show clear and smooth periodic oscillations across interaction length an polarization state. The consistent and continuous periodicity suggesting a stable coupling effect. Although there are few irregularities and noise, this can be attributed to experimental noise and practical limitations in the setup.

Aplitude Range The numerical range in the theoretical model spans from 0 to 1, while the numerical range of the experimental data spans from about 0.2 to 0.8, which is close to that of the theoretical model, indicating that the experimental data are on the same scale as the theoretical predictions.

Phase Shift The experimental data shows slight shifts and distortions. This might be the reason of effective interaction length as the waveguides might start talking to each other earlier than what we expect.

MSE Minimization By observing the plot, the experiment shows good agreement with the theoretical prediction. However, we still need compensation for the assumptions made and the imperfect reality. Therefore, mean squared error (MSE) is needed to provide a quantified comparison between the two.

Model Compensation: In this case, the phase shift in interaction length, magnitude reduction and the environment noise are obviously needed to be modified. However, simultaneous modification of three variables are hard to begin with. Hence we would like to rely on the quantification method which is to minimize the MSE between the theory model and experimental results.

We used the L-BFGS-B algorithm, which is effective for optimizing parameters within specified bounds. The parameters optimized included the wave vector components k_x and k_y , the phase shift (*shift*), and the ambient power:

- k_x (2.5-3.5mm)
- k_y (2.5-3.5mm)
- *shift* (0-2mm)
- Ambient power (0-0.3 units)

The modified theoretical model including ambient power is given by:

$$r(L_0, \theta) = M \left(\sin^2 \left(\frac{k_x}{2} (L_0 + \text{shift}) \right) + \sin^2(\theta_{\text{rad}}) \left(\sin^2 \left(\frac{k_y}{2} (L_0 + \text{shift}) \right) - \sin^2 \left(\frac{k_x}{2} (L_0 + \text{shift}) \right) \right) \right) + \text{ambient power}$$

The optimization results are as follows:

- Optimal k_x : 3.004
- Optimal k_y : 2.89
- Optimal *shift*: 0.512
- Optimal ambient power: 0.206
- Minimum MSE: 0.0087(0.87 percent)

Hence the model after optimization is:

$$r(L_0, \theta) = 0.75 \left(\sin^2 \left(\frac{k_x}{2} (L_0 + 0.512) \right) + \sin^2(\theta_{\text{rad}}) \left(\sin^2 \left(\frac{k_y}{2} (L_0 + 0.512) \right) - \sin^2 \left(\frac{k_x}{2} (L_0 + 0.512) \right) \right) \right) + 0.206$$

where k_x and k_y are now 3.004mm^{-1} and 2.89mm^{-1}

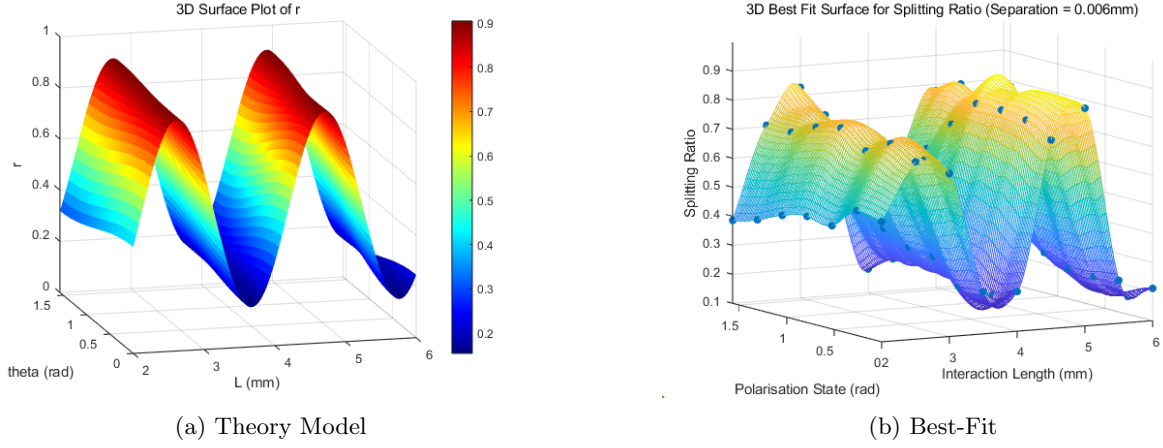


Figure 25: Comparison between Theory Model with compensation and Best-fit

0.87% MSE is really low which perfectly indicate 0.006mm separation coupler has great quality and precision in prediction. It also shows a consistent and stable coupling effect.

Choosing Suitable Interaction Length for Polarimeter Use: For separation=0.006mm, the most suitable interaction length in the range 2mm-6mm is 3.5mm which has greatest slope and significant change with the variation of polarization state.

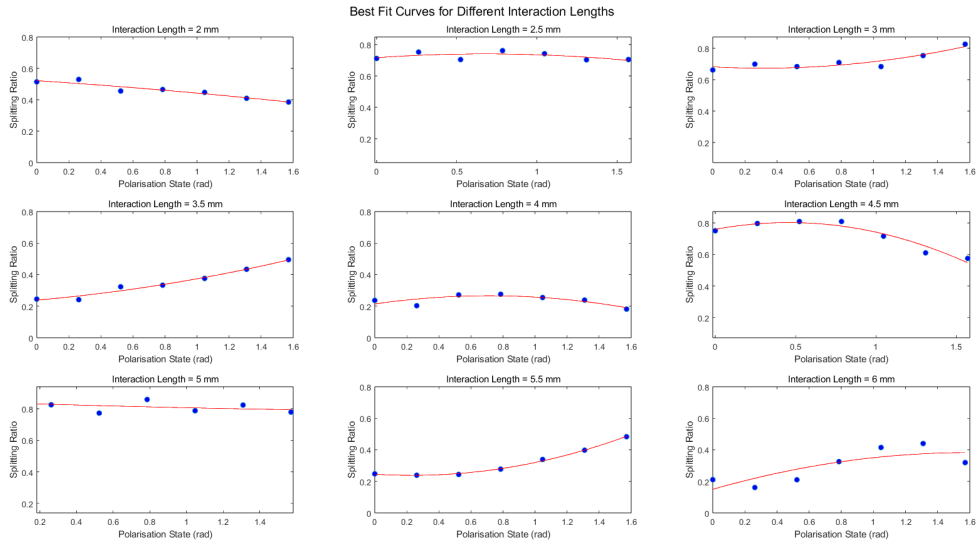


Figure 26: Splitting Ratio vs. polarization at different interaction length (Experiment)

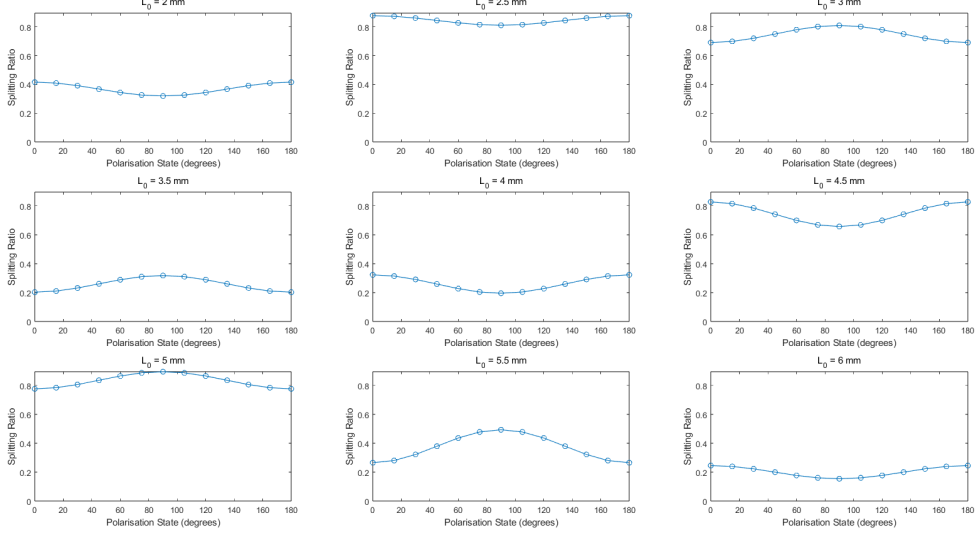


Figure 27: Splitting Ratio vs. polarization at different interaction length (Compensated Theory)

2.5 Traditional Coupler as Polarimeter

2.5.1 Coupler Characteristic Design

One appropriate choice of separation and interaction length is 0.008mm separation with 9mm interaction length, which has great change of splitting ratio with change of polarization. Moreover, 8.5mm and 9.5mm also have significant change so that if there is any change of effective interaction length, the polarimeter will still have good performance. However, in case there is shift in effective interaction length, separation 8mm and 10mm are also fabricated to avoid it. By observing the changes over polarization state, 10mm perform much better than 9mm, hence we decide to choose 10mm coupler as the polarimeter.

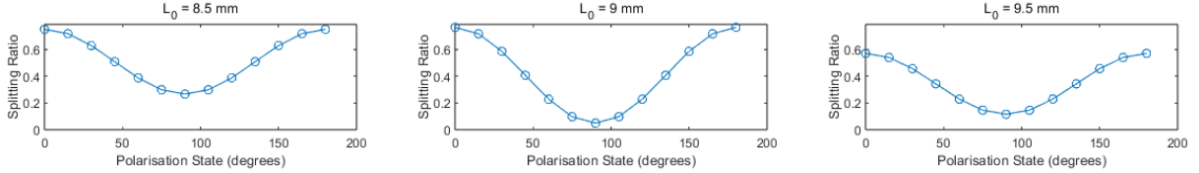


Figure 28: The Designed Interaction Length for Polarimeter Use

2.5.2 Fabrication

The fabrication process and the Aerotech-code are the same as the previous traditional coupler's sample. As the desired designs are for 8mm, 9mm and 10mm, we still need to fabricate samples with same separation but different interaction length from 4mm-11mm to estimate the k_x and k_y . They are supposed to be the same as the previous 0.008mm-separation sample.

2.5.3 Testing and Calibration

Testing The estimation from the experimental data with 0 degree and 90 degree polarization state can be used to calculate the k_x and k_y . By plot the splitting ratio over interaction length, k_x and k_y are estimated to $1.4568mm^{-1}$ and $1.6456mm^{-1}$.

Then we test the splitting ratio with few different polarization state from 0 degree, 10 degrees, 20 degrees and so on to calibrate the coupler which is basically compensation part. The compensated model is given by:

$$M \left(\sin^2 \left(\frac{k_x}{2} (L_0 + shift) \right) + \sin^2(\theta_{rad} + angleshift) \left(\sin^2 \left(\frac{k_y}{2} (L_0 + shift) \right) - \sin^2 \left(\frac{k_x}{2} (L_0 + shift) \right) \right) \right) + \text{ambient power}$$

Calibration We used the L-BFGS-B algorithm then. The parameters optimized included the wave vector components k_x and k_y , the interaction length shift ($shift$), the angle shift($angleshift$), and the ambient power:

- k_x (1.4-1.5mm)
- k_y (1.6-1.7mm)
- $shift$ (0-2mm)
- Ambient power (0-0.1 units)
- $angleshift$ (0-20 degrees)
- M: 0.9

The optimization results are as follows:

- Optimal k_x : 1.454
- Optimal k_y : 1.673
- Optimal $shift$: 0.975
- Optimal $angleshift$:9.20
- Optimal ambient power: 0.045
- Minimum MSE: 0.00018(0.018 Percent)

Hence the model after optimization is:

$$0.9 \left(\sin^2 \left(\frac{1.454}{2}(10 + 0.975) \right) + \sin^2(\theta_{\text{rad}} + 9.20) \left(\sin^2 \left(\frac{1.673}{2}(L_0 + 0.975) \right) - \sin^2 \left(\frac{1.454}{2}(10 + 0.975) \right) \right) \right) + 0.045$$

2.5.4 Polarization Measurement

We could produce a list of splitting ratio predict based on the optimization model we got after calibration. Hence the predict values can be then compared with the actual value measured.

Polarisation state	Splitting Ratio Predict	Splitting Ratio Measured
0	0.88964	0.902654867
10	0.928943	0.91
20	0.918237	0.909774436
30	0.858813	0.865168539
40	0.75784	0.76
50	0.627495	0.629139073
60	0.4835	0.51369863
70	0.343223	0.352941176
80	0.223584	0.209580838
90	0.139013	0.129411765
100	0.09971	0.087248322
110	0.110416	0.102702703
120	0.16984	0.198041349
130	0.270814	0.277777778
140	0.401159	0.398734177
150	0.545154	0.538461538
160	0.68543	0.662420382
170	0.805069	0.808510638
180	0.88964	0.896774194

Figure 29: Splitting Ratio Predicted and Measured

The calculated mean difference between the predicted and measured Splitting Ratios is -0.000285, which is very close to zero. This indicates that, overall, the values in the two datasets are very similar. Additionally, the standard deviation of the differences is 0.013855, which is quite small. This suggests that the differences across various polarization angles are minimal. Therefore, we can conclude that the predicted and measured Splitting Ratios are almost identical, demonstrating a high degree of consistency between the model's predictions and the actual measurements.

The predicted value and measured value are the same and distinguishable, by achieving this, we could then predict the polarization state angle based on the splitting ratio measured as input variables from the predict model. However, the angle measured has two possible solutions, which are θ itself and $(\theta + 90)$. To further identify and determine the exact angle, we need further information on the phase difference between the x and y component of the E field.

2.6 Polarization Independence at Certain Interaction Length

In most cases, we do not want the interaction length to be too long. Simply design is always preferred. Hence we need to investigate how to make the polarization-independent interaction length small. Consider the mathematical model for the splitting ratio:

$$\text{Splitting Ratio} = \sin^2 \left(\frac{k_x L_0}{2} \right) + \sin^2 \theta \left(\sin^2 \left(\frac{k_y L_0}{2} \right) - \sin^2 \left(\frac{k_x L_0}{2} \right) \right)$$

1. Condition for Independence from Polarization Angle

The splitting ratio will be independent of the polarization angle θ when the second term becomes zero, i.e.,

$$\sin^2 \theta \left(\sin^2 \left(\frac{k_y L_0}{2} \right) - \sin^2 \left(\frac{k_x L_0}{2} \right) \right) = 0$$

When $\sin^2 \left(\frac{k_y L_0}{2} \right) = \sin^2 \left(\frac{k_x L_0}{2} \right)$, the splitting ratio becomes independent of θ . This implies:

$$\frac{k_y L_0}{2} = \frac{k_x L_0}{2} + n\pi \quad \text{for some integer } n$$

For the simplest case $n = 0$, this means $k_y = k_x$, making the splitting ratio independent of θ for all L_0 .

2. Ensuring Interaction Length $L_0 < 10$ mm

To ensure $L_0 < 10$ mm, we need to satisfy the above conditions and choose small values of L_0 .

Since the sine function repeats every 2π , we should consider the first condition, $k_y = k_x$, and examine specific values of k_x and k_y that lead to small L_0 .

$$L_0 = \frac{2n\pi}{|k_x - k_y|}$$

For $L_0 < 10$ mm, the values of k_x and k_y must be such that the difference $|k_x - k_y|$ is sufficiently large to ensure a small L_0 .

3. Special Case: $k_x \approx k_y$

In practice, if k_x and k_y are nearly equal, the polarization dependence will be minimized, and the splitting ratio will remain approximately constant over small L_0 .

Specific values can be calculated by ensuring $L_0 < 10$ mm based on the relationship above.

Summary

The interaction length that present polarization independence can be calculated from

$$L_0 = \frac{2n\pi}{|k_x - k_y|}$$

To make the desired interaction length small, the difference between the k_x and k_y has to be large or exactly the same. Moreover, we already observe that the closer separation distance is, the bigger coupling coefficient difference will be. Hence we need to fabricate small separation spacing to make sure the polarization independent interaction length small.

2.7 Conclusion

In this work, we have demonstrated that by knowing the x and y components of the polarization state, we can narrow down the polarization to two possible states. However, to precisely identify the exact polarization state, additional information on the phase difference between the x and y components is necessary. This can be achieved in future work by incorporating a waveplate into the setup, which will allow for the determination of the phase difference.

Furthermore, we have established that for any given separation and interaction length, a polarimeter can be effectively designed, except in cases where the interaction length is polarization-independent. This polarization-independent interaction length can be predicted and calculated in advance, allowing us to either avoid or deliberately apply it depending on the desired application.

Lastly, we have confirmed the accuracy of the assumption that the LP01 mode can be considered as a weighted sum of the x-mode and y-mode, or equivalently the TE mode and TM mode. This assumption holds true under weakly guiding conditions and within the LP approximation, validating our approach and providing a solid foundation for further research and practical applications.

3 Investigate Four Waveguides Coupling with center waveguide

3.1 Separate Coupling

3.1.1 Structure Design

The five waveguides are not coupled together but separate to horizontal three waveguides coupler and vertical three waveguides coupler. The horizontal one is designed to be the first coupling. Each coupling

has all the same setting like separation distance or interaction length but coupled at different positions. However, the separation now has doubled between the coupler, which is 0.016mm between the horizontal coupler and 0.008mm between two adjacent waveguides.

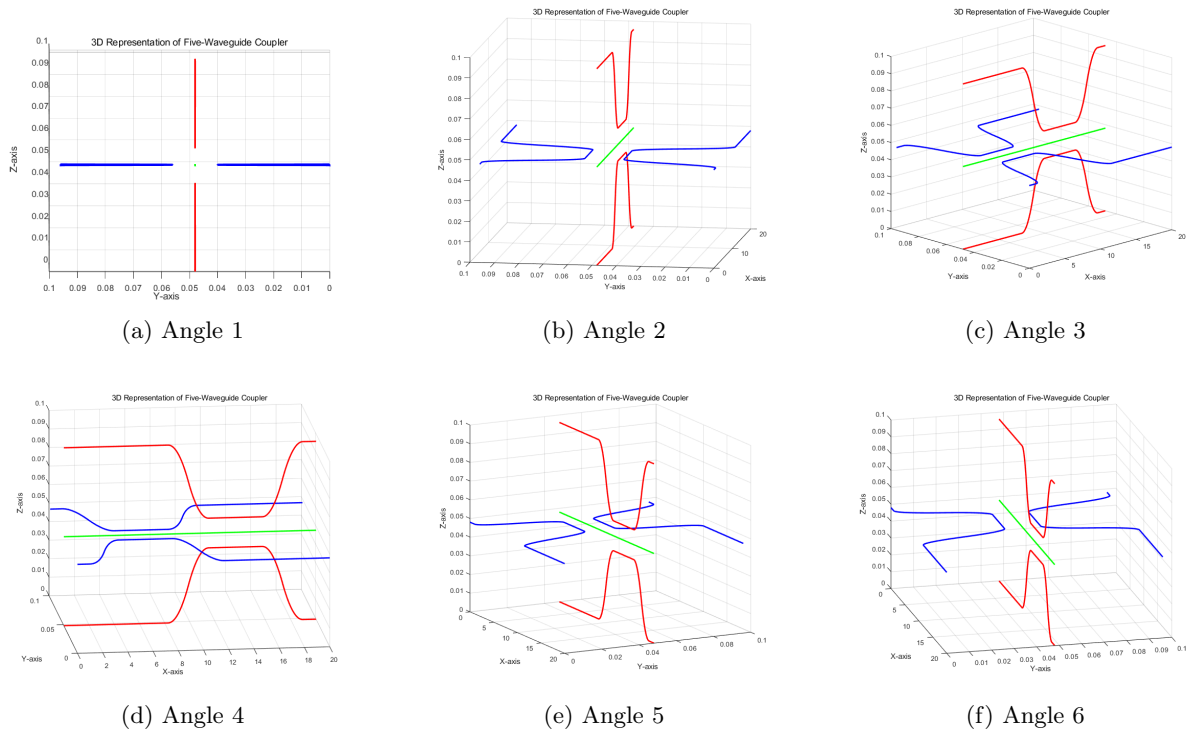
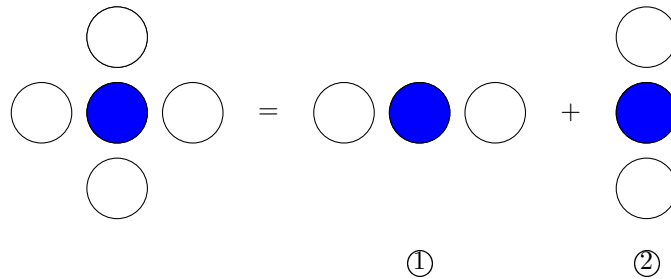
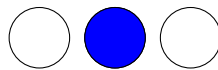


Figure 30: 3D Representation of Five-Waveguide Coupler from Different Angles

3.1.2 Theory Model



For the horizontal planar Three-Waveguide Coupler:



The Coupled Mode Equations of the System are the followings [9].

$$\frac{da_1}{dz} = jka_2 \quad (32)$$

$$\frac{da_2}{dz} = jka_1 + jka_3 \quad (33)$$

$$\frac{da_3}{dz} = jka_2 \quad (34)$$

We could express the Coupled Mode Equation in the matrix form.

$$\frac{d}{dz} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 0 & jk & 0 \\ jk & 0 & jk \\ 0 & jk & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \quad (35)$$

Find eigenvalues and corresponding eigenvectors to determine the eigensolutions of three-guide system:

$$\begin{vmatrix} -\lambda & jk & 0 \\ jk & -\lambda & jk \\ 0 & jk & -\lambda \end{vmatrix} = 0 \quad (36)$$

$$\lambda = 0, \pm\sqrt{2}k \quad \text{and} \quad v = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix} \quad (37)$$

The eigensolutions can be expressed as the linear combination of eigenvectors.

$$\vec{a}(z) = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = C_1 v_1 e^{\lambda_1 z} + C_2 v_2 e^{\lambda_2 z} + C_3 v_3 e^{\lambda_3 z} \quad (38)$$

Our design for the Five-WGs coupler has input light at the center waveguide. Hence we have the initial condition:

$$\vec{a}(0) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = C_1 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + C_2 \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix} + C_3 \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix} \quad (39)$$

C_1 , C_2 and C_3 can then be solved:

$$C_1 = 0 \quad (40)$$

$$C_2 = \frac{1}{2\sqrt{2}} \quad (41)$$

$$C_3 = \frac{1}{2\sqrt{2}} \quad (42)$$

Now we obtain the solutions for individual waveguide in this planar system:

$$a_1(z) = \frac{1}{2\sqrt{2}} \left(e^{j\sqrt{2}kz} - e^{-j\sqrt{2}kz} \right) = \frac{1}{\sqrt{2}} \sin(\sqrt{2}kz) \quad (43)$$

$$a_2(z) = \frac{1}{2} \left(e^{j\sqrt{2}kz} + e^{-j\sqrt{2}kz} \right) = \cos(\sqrt{2}kz) \quad (44)$$

$$a_3(z) = \frac{1}{2\sqrt{2}} \left(e^{j\sqrt{2}kz} - e^{-j\sqrt{2}kz} \right) = \frac{1}{\sqrt{2}} \sin(\sqrt{2}kz) \quad (45)$$

where k could be k_x and k_y at the same time

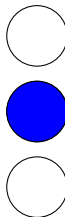
The power at the out-ports after interaction length L_0 can be now found:

$$P_1 = \frac{1}{2} \sin^2 \left(\frac{k_x}{2} L_0 \right) \cos^2 \theta + \frac{1}{2} \sin^2 \left(\frac{k_y}{2} L_0 \right) \sin^2 \theta \quad (46)$$

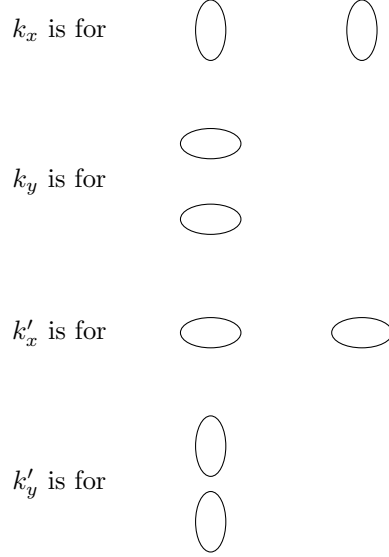
$$P_2 = \cos^2 \left(\frac{k_x}{2} L_0 \right) \sin^2 \theta + \cos^2 \left(\frac{k_y}{2} L_0 \right) \cos^2 \theta \quad (47)$$

$$P_3 = \frac{1}{2} \sin^2 \left(\frac{k_x}{2} L_0 \right) \cos^2 \theta + \frac{1}{2} \sin^2 \left(\frac{k_y}{2} L_0 \right) \sin^2 \theta \quad (48)$$

For the vertical planar Three-Waveguide Coupler:



The k_x and k_y exchange their positions as the relative polarization state has rotated 90 degrees. However, the new k_x is different from the horizontal one because of the birefringence effect, as well as k_y . Moreover, previously we used the unit input power but now the input power to the vertical planar coupler is the power out-port of the center waveguide of the horizontal planar coupler which is P_2 .



Hence the power out-ports of vertical waveguides are:

$$P'_1 = P_2 \left[\frac{1}{2} \sin^2 \left(\frac{k'_x}{2} L_0 \right) \sin^2 \theta + \frac{1}{2} \sin^2 \left(\frac{k'_y}{2} L_0 \right) \cos^2 \theta \right] \quad (49)$$

$$P'_2 = P_2 \left[\cos^2 \left(\frac{k'_x}{2} L_0 \right) \sin^2 \theta + \cos^2 \left(\frac{k'_y}{2} L_0 \right) \cos^2 \theta \right] \quad (50)$$

$$P'_3 = P_2 \left[\frac{1}{2} \sin^2 \left(\frac{k'_x}{2} L_0 \right) \sin^2 \theta + \frac{1}{2} \sin^2 \left(\frac{k'_y}{2} L_0 \right) \cos^2 \theta \right] \quad (51)$$

However, we could only measure P'_1 , P'_2 , P'_3 , P_1 and P_3 at the end. Hence there will be two definition of splitting ratio.

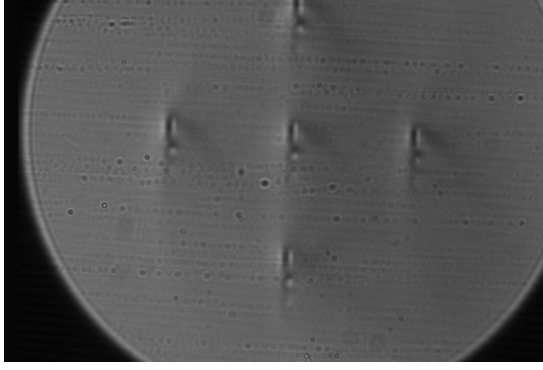
$$r_1 = \frac{P'_1}{P'_1 + P'_2 + P'_3} = \frac{1}{2} \sin^2 \left(\frac{k'_x}{2} L_0 \right) \sin^2 \theta + \frac{1}{2} \sin^2 \left(\frac{k'_y}{2} L_0 \right) \cos^2 \theta \quad (52)$$

$$r_2 = \frac{P_1}{P_1 + P_3 + P'_1 + P'_2 + P'_3} = \frac{1}{2} \sin^2 \left(\frac{k_x}{2} L_0 \right) \cos^2 \theta + \frac{1}{2} \sin^2 \left(\frac{k_y}{2} L_0 \right) \sin^2 \theta \quad (53)$$

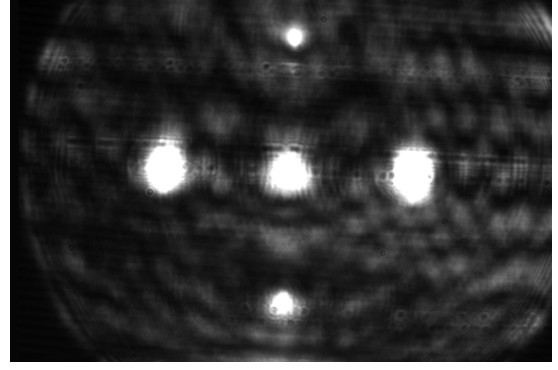
3.1.3 Fabrication Process

The Aerotech G-code is generated by the MATLAB code. Hence, to write the fabrication code for five-WGs coupler, we just need to write the two coupler twice but change the interaction position and most importantly exchange the y and z coordinates. Thus the vertical coupler can be duplicated by copying the horizontal coupler's code. Then write a straight waveguide at their center.

However, to reduce the complexity of writing the script and difficult of fabrication, we decided to ignore the small offset of the chip. This leads to little deformation of the laser focal shape which might cause imperfect waveguides and coupling as well. It can be seen in the image below that the waveguide cross-section shape is bit elongated. The specific influence has to be confirmed in the experimental results.



(a) Five-WGs captured on camera with LED light



(b) Five-WGs captured on camera with laser

Figure 31: Five-WGs Captured on Camera

3.1.4 Experimental Results

The theory looks promising but the experimental data presents non-expected results and unknown randomness.

IX	Left	Centre	Right	Up	Bottom	r1	r2
2mm (Polarization 0)	0.053	0.414	0.045	0.365	0.872	0.221078	0.030303
2.5mm (Polarization 0)	0.219	0.387	0.211	0.469	0.473	0.352897	0.124503
3mm (Polarization 0)	0.443	0.281	0.382	0.058	0.144	0.120083	0.338685
3.5mm (Polarization 0)	0.459	0.072	0.465	0.045	0.039	0.288462	0.425000
4mm (Polarization 0)	0.295	0.144	0.343	0.089	0.122	0.250704	0.297080
2mm (Polarization 15)	0.175	0.795	0.180	0.630	1.720	0.200318	0.050000
2.5mm (Polarization 15)	0.430	1.155	0.520	0.845	1.340	0.252994	0.100233
3mm (Polarization 15)	0.900	0.715	0.825	0.070	0.345	0.061947	0.315236
3.5mm (Polarization 15)	0.745	0.160	0.815	0.070	0.140	0.189189	0.386010
4mm (Polarization 15)	0.650	0.230	0.800	0.200	0.230	0.303030	0.308057
2mm (Polarization 30)	0.610	0.420	0.710	0.150	0.240	0.185185	0.286385
2.5mm (Polarization 30)	0.550	0.140	0.710	0.050	0.080	0.185185	0.359477
3mm (Polarization 30)	0.620	0.460	0.610	0.100	0.090	0.153846	0.329787
3.5mm (Polarization 30)	0.390	0.560	0.400	0.630	0.480	0.377246	0.158537
4mm (Polarization 30)	0.120	0.110	0.110	0.360	0.540	0.356436	0.096774

As we observe from the table, the left-right and up-bottom are not symmetrical as they are always different. Especially up and bottom measurements, the difference could become significantly large sometimes. Hence we could conclude that our assumption of the theory has failed. It may be caused by imperfect fabrication that one of the waveguide in the symmetric coupler talk to the centre one first and take away the majority of power. Or it may be caused by the wrong predicted model but I think this is not much possible. The predicted model should work if the sample is perfectly fabricated and truly symmetrical.

As the left-right and up-bottom are different, we should define two more splitting ratio to count for power at right waveguide and bottom waveguide.

IX	r3	r4
2mm (Polarization 0)	0.528165	0.025729
2.5mm (Polarization 0)	0.355907	0.119955
3mm (Polarization 0)	0.298137	0.292049
3.5mm (Polarization 0)	0.250000	0.430556
4mm (Polarization 0)	0.343662	0.345418
2mm (Polarization 15)	0.546900	0.051429
2.5mm (Polarization 15)	0.401198	0.121212
3mm (Polarization 15)	0.305310	0.288967
3.5mm (Polarization 15)	0.378378	0.422280
4mm (Polarization 15)	0.348485	0.379147
2mm (Polarization 30)	0.296296	0.333333
2.5mm (Polarization 30)	0.296296	0.464052
3mm (Polarization 30)	0.138462	0.324468
3.5mm (Polarization 30)	0.287425	0.162602
4mm (Polarization 30)	0.534653	0.088710

3.1.5 Conclusion

The measurement data exhibit a somewhat complex pattern, which could be attributed to sensitivity to fabrication imperfections or potentially an underlying theoretical model. It is not immediately clear if there is a theory that can fully describe this scenario, so more precise and careful measurements and fabrications are needed to further investigate the Five-WGs coupler.

4 Further Possible Exploration

4.1 Influence of Traditional Coupler to Out-port Polarization State

Traditional Coupler is originally not to alter the polarization state of the input light. However, under some conditions like birefringent coupler, the coupler might look like a waveplate introducing phase shift to different polarization component.

4.2 Influence of Circular Polarization State to Splitting Ratio

So far, the laser we have been using only emits light with a linear polarization state. However, when a circular polarization component is present, the situation may change. Under ideal weakly guiding conditions, the longitudinal field components are typically ignored. As a result, the LP approximation remains valid, along with the weighting method, because we can still consider the circular waveguide as analogous to a slab waveguide. However, when the circular component dominates, the hybrid mode (HE mode) will inevitably arise. This could undermine the idealization of the slab waveguide. Therefore, it is necessary to investigate the influence of a circular polarization state on the validity of the LP approximation.

4.3 Investigate the Validity of the Weighted Sum of modes(x and y mode) Assumption when more complex modes are present

By increase the V number, we excite more complex modes and higher order modes. Hence our assumption that the power can be seen as the weighted sum of power of x-mode and y-mode might fail. Then the theory model might not be able to predict the behavior well. Just remember, the weakly guiding has to be achieved, which provide a chance to consider circular waveguide as slab waveguide.

Consider the conditions under which the theoretical model applies. If the experimental data is limited to a specific parameter range (e.g., wavelength, waveguide shape, waveguide size), a good fit may only hold within these conditions. Try extending the experimental parameters to see if the weighting approach still works.

4.4 Measure the Five-WGs Separate Coupling Coupler Again with Higher-precision measuring instruments

The Five-WGs coupler is more sensitive to the optical fiber alignment, distance between optical fiber and waveguide in-ports and fabrication errors. Hence we need to treat the offset observed in fabrication process carefully to make sure the laser focal has no deformation which affects the experimental results significantly.

4.5 Fabricate the Five-WGs Coupler that coupled together

The Five-Waveguide (WGs) Coupler, which couples multiple waveguides together, has not yet been fabricated or thoroughly studied. The coupled modes within this structure will likely be much more complex, and there is currently limited research available on spatial coupling in such configurations. However, by simply observing the patterns that emerge, we may be able to extract valuable information that could guide the development of a theoretical framework for this coupler.

For an ideal coupler with no birefringence effects, the four waveguides surrounding the central one should behave symmetrically. However, birefringence in the horizontal and vertical directions may lead to unique phenomena that are sensitive to the polarization state of the light. This is a plausible hypothesis, but it needs to be experimentally confirmed to determine its usefulness. Alternatively, a Five-Waveguide Coupler with surrounding waveguides that are not symmetrically aligned but are instead positioned with a designed offset could be explored as another option.

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